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SUPERSEDING
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17 March 1975
(See 6.11)

MILITARY SPECIFICATION

CANISTER, C-1

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a canister for use with the Navy Mark V protective mask for respiratory protection against chemical, biological, and radiological warfare agents.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-C-96	- Cans, Metal, 28 Gage and Lighter.

MILITARY

MIL-C-13724	- Charcoal, Activated, Impregnated, ASC.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-313	- Material Safety Data Sheets Preparation and the Submission of.
FED-STD-595	- Colors.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

2.1.2 Government drawings. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

803-1327146	- Mask, Chem-Biol, MK V C-1 Canister Assembly and Details.

US ARMY MUNITIONS AND CHEMICAL COMMAND CHEMICAL RESEARCH AND DEVELOPMENT CENTER

C26-19-51	- Machine, Rough Handling, Q113.
E136-13-362	- Test Device, Pull, Canister Apron, Q22.
E136-16-3	- Meter, Volume, Chemical Containers, Q23.
DLB31-6-351	- Indicator, Resistance, Canister, Air Purifier and Chemical Container, Q112.
DLB136-41-600	- Apparatus, All Purpose Gas Life Testing, Q95.
DLB136-42-102	- Penetrometer, Filter Testing, DOP, Q127.
DLB136-41-1800-5	- Tester, Filter Life, 50 LPM, Q224.
PDS-5-3-578	- Packaging Data Sheet.

(Copies of drawings may be obtained from the US Army Armament Munitions and Chemical Command Chemical Research and Development Center, Documentation Center, Code AMSMC-TSC Aberdeen Proving Ground, Edgewood, MD 21010-5423.)

NAVAL SURFACE WEAPONS CENTER

07609-65836	- Handling Device, Rough, NRL.

(Copies of drawings may be obtained from Naval Surface Weapons Center, Documentation Center, Code E231D, Dahlgren, VA 22448.)

(Copies of specifications, standards, handbooks, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 395 - Rubber Property - Compression Set, Test Methods for. (DoD adopted)
D 412 - Rubber Properties in Tension, Test Methods for. (DoD adopted)
D 573 - Rubber - Deterioration in an Air Oven, Test Method for. (DoD adopted)
D 1149 - Rubber Deterioration - Surface Ozone Cracking in a Chamber (Flat Specimen), Test Method for. (DoD adopted)
D 1415 - Rubber Property - International Hardness, Test Method for. (DoD adopted)
D 2867 - Moisture in Activated Carbon, Test Method for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

UNIFORM CLASSIFICATION COMMITTEE AGENT

Uniform Freight Classification Ratings, Rules and Regulations

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Materials. Materials shall be in accordance with Drawing 803-1327146 and as specified herein.

3.2.1 Rubber compound. The rubber compound shall be non-sulphurblooming and contain not less than 85 percent by volume of natural rubber, conforming to the requirements shown in table I. The rubber compound shall contain less than 1 part per million (0.0001 percent) of mineral sulfates (lead, calcium, magnesium, potassium, or sodium) and contain not more than 3.5 percent sulphur by weight. The rubber of the finished parts shall have the physical characteristics specified in table I.

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TABLE I. Physical characteristics of rubber compound.

Properties	Requirement
Initial properties	
Tensile strength minimum (lb/in^2)	3000
Elongation (minimum percent)	650
Tension set (maximum percent)	20
Hardness-Shore A	45 ± 5
Tensile stress at 200 percent elongation (lb/in^2)	200-400
Properties after aging	
Tensile strength minimum (lb/in^2)	2250
Elongation (minimum percent)	650 ± 20 percent
Tensile stress at 200 percent elongation (lb/in^2)	200-400
Compression set (maximum percent)	45

3.2.1.1 Sun checking compound. The rubber compound shall contain from 0.5 to 1.0 percent by weight a sun checking compound (additive to block out ultra-violet rays) known commercially as Wito Company Sunolite Number 127 with Uniroyal Corporation Flexzone 3C or Government approved equivalent (see 6.10).

3.2.1.2 Ozone resistance. The rubber compound shall contain from 1.5 to 2.5 percent by weight an antioxidant compound which can pass the test of 4.6.1.2.7.

3.2.1.3 Mildew inhibitor. The rubber compound shall contain from 1.0 to 2.0 percent by weight, a non-mercury containing mildew inhibitor that will not adversely affect the physical properties of the rubber listed in table I, such as those known commercially as Vanderbilt Corporation Agerite White or Government approved equivalent (see 6.10).

3.2.1.4 Color. The rubber compound shall be colored a dull black to conform to FED-STD-595, color number 37038, by mineral pigments or carbon blacks.

3.2.1.5 Toxicity. The rubber apron shall have no adverse effect on the health of personnel when properly used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting agency.

3.2.2 Particulate filter medium. Binding agents and other additives may be incorporated, provided that the combined weight of all additives, including inhibitors and waterproofing, does not exceed 5 percent of the total weight of the filter. Waterproofing may be accomplished through the use of an appropriate additive or by a surface treatment of the completed filter.

3.2.2.1 Filter paper additives. Inhibitors and waterproofing additives shall show no deleterious effects (will not adversely affect the life and utility of the canister) on the performance requirements of the completed canister when tested in accordance with 4.6.2.8.

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3.2.3 Cloth disk. The cotton flannel disk on Drawing 803-1327146 shall not have a resistance to air flow greater than 1.5 millimeters (mm) water when tested in accordance with 4.6.1.3.

3.2.4 Adsorbent material (charcoal). The adsorbent material in the canister shall meet the requirements of grade II of MIL-C-13724. The adsorbent in the completed canister shall completely pass a U.S. Standard No. 10 sieve, shall not contain in excess of 0.5 percent by weight of material which will be retained by a U.S. Standard No. 12 sieve, and shall not contain in excess of 2.0 percent by weight of material which will pass a U.S. Standard No. 30 sieve. Sieve size determinations shall be made in accordance with 4.6.1.4.2. The volume shall be 158 + 2, - 4 cubic centimeters when tested in accordance with 4.6.1.4.1. Precautions shall be taken to ensure that the adsorbent material is kept free of foreign matter and not exposed to atmospheric moisture.

3.2.5 Material safety data sheet. The contracting activity shall be provided a completed material safety data sheet (MSDS) at the time of contract award by the contractor. The MSDS is form OSHA-20, found in and part of FED-STD-313. The MSDS shall be included with each shipment of the material covered by this specification.

3.3 Manufacture and assembly.

3.3.1 Type. The canister shall be manufactured in accordance with Drawing 803-1327146 and designated as "C-1".

3.3.2 Assembly filling and closing of canister. The components of the canister shall be assembled as shown on Drawing 803-1327146.

3.4 Canister performance.

3.4.1 Air flow resistance. The resistance of each canister to air flow shall be not less than 47 mm, nor greater than 67 mm of water, when determined in accordance with 4.6.2.8. No two canisters having a resistance differential greater than 8 mm of water shall be packaged in the same container.

3.4.2 Aerosol penetrations. The penetration of each canister by a homogeneous aerosol of DOP of 0.3 micrometer diameter shall not exceed 0.015 percent when tested in accordance with 4.6.2.1, using DOP smoke at a concentration of 100 to 125 micrograms per liter and a flow rate of 16 liters per minute (L/min).

3.4.3 Aerosol penetration after rough handling. After rough handling as specified in 4.6.2.2 and 4.6.2.3, the DOP penetration of each canister shall not exceed 0.015 percent when tested in accordance with 4.6.2.1.

3.4.4 Gas life.

3.4.4.1 Cyanogen chloride (CK). The cyanogen chloride computed gas life, mean minus 1.1 times the standard deviation (see 6.7), shall be no less than 7.0 minutes when corrected to a challenge concentration of 4000 milligrams per cubic meter (mg/m^3) when tested as specified in 4.6.2.4.1. Cyanogen chloride gas life is defined as the time required to attain an effluent cyanogen chloride concentration of 8 mg/m^3 when the canister is challenged as follows:

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(a) Agent: cyanogen chloride, technical grade (ClCN).
(b) Concentration: 4000 ± 200 mg/m^3.
(c) Flow: 25.0 ± 0.8 L/min, intermittent 33-1/2 cycles per minute.
(d) Temperature: 80 ± 10°F.
(e) Relative humidity: 80 ± 3 percent.

3.4.4.2 Sarin (GB) gas life. The GB computed gas life, mean minus 1.1 times the standard deviation, shall be no less than 40.0 minutes when corrected to a challenge concentration of 4000 mg/m^3 when tested as specified in 4.6.2.4.2. GB gas life is defined as the time needed to attain an effluent GB concentration of 0.04 mg/m^3 when the canister is challenged as follows:

(a) Agent: GB, purity 96 percent minimum.
(b) Concentration: 4000 ± 400 mg/m^3.
(c) Flow: 25.0 ± 0.8 L/min, continuous.
(d) Temperature: 75 ± 5°F.
(e) Relative humidity: not more than 15 percent.

3.4.4.3 GB gas life after rough handling. After rough handling for 10 minutes in accordance with 4.6.2.4.3, the computed GB gas life, mean minus 1.1 times the standard deviation, shall be no less than 40.0 minutes when tested as specified in 4.6.2.4.2.

3.4.5 Apron staking. The canister apron shall not pull out under a load of 25 pounds when tested in accordance with 4.6.2.5.

3.4.6 Volatile content of the canister. The volatile matter in the completed canister, based on the dry weight of the charcoal as determined by 4.6.2.6, shall not exceed 1.8 percent by weight.

3.4.7 Performance after aging. After the canister is aged in accordance with 4.6.2.7, it shall show no degradation from the DOP penetration and mean gas life of an unaged canister when tested in accordance with 4.6.2.1, 4.6.2.4.1 and 4.6.2.4.2.

3.5 Marking. Marking of the canister shall be in accordance with Drawing 803-1327146.

3.6 Workmanship. The particulate filter units shall be free from foreign matter, wrinkles, tears, punctures, pickouts, fractures, or holes. The canister shall be free from contamination such as dirt, grease, oil, foreign matter, and damage such as cracks, dents, burrs, tears, punctures, scratches, and rust. The rubber surfaces shall be free of pits, bumps, nonfills, porous areas, spongy areas, blisters, air traps, deformation, flow cracks, foreign matter, flash abrasions, and damage such as tears, rips, cuts, holes, or punctures. Ridges of mold parting lines on outer surfaces shall not be considered flash when closely trimmed.

3.7 Control of manufacture. The contractor shall provide the Government with the exact (not typical) procedures, process controls, and conditions and their limits to produce canister under this specification (see 6.2.2).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The contractor shall provide and maintain an inspection system in accordance with the data ordering documents included in the contract or order (see 6.2.2).

4.1.2 Toxicity report. The contractor shall prepare a toxicity report for each lot of rubber in accordance with the data ordering document included in the contract (see 6.2.2). In addition to the requirements covered by the data ordering document, the toxicity report shall include the following information:

(a) Toxicological data and formulations required to evaluate the safety of the material for the intended use (see 3.2.1.5).
(b) Certified data that volatility requirements are met (see 3.4.6).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection.

4.3.1 Lot. Prior to the start of regular production and following any change in regular production methods, materials, or equipment, a first article lot of 250 hermetically sealed containers each containing two complete canisters, 50 particulate filter media disks, and 50 of each of the other canister parts exclusive of the adsorbent, shall be produced using the same methods, materials, and equipment as used during regular production.

4.3.2 First article inspection procedure.

4.3.2.1 For examination and nondestructive tests. Fifty hermetically sealed containers, each containing two complete canisters, randomly selected from the first article lot; 50 each particulate filter media disks; and 50 each of the other canister components shall be inspected and tested for compliance with all requirements of Drawing 803-1327146 and 4.5.

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4.3.2.2 For destructive tests. Representative canisters shall be taken at random from the first article lot and subjected to the tests specified in table II. Seventy-five containers, each containing two canisters, shall be randomly selected from each lot. From these 75 containers, 25 containers suitably marked, shall be subjected to the aging test in 4.6.2.7. All 75 containers (150 canisters) shall be forwarded to the Government test agency as specified in the contract or order (see 6.2.1) for GB and CK gas life tests specified in 4.6.2.4.

TABLE II. First article inspection (destructive tests).

Item or characteristics	Requirement paragraph	Test paragraph	Number of canisters before rough handling	Number of canisters after rough handling	Number of individual canister parts
Particulate filter medium	3.2.2	4.6.1.1	---	---	50
Cloth disk	3.2.3	4.6.1.3	---	---	50
Adsorbent sieve size	3.2.4	4.6.1.4.2	16	---	---
Adsorbent volume	3.2.4	4.6.1.4.1	16	---	---
Canister apron staking	3.4.5	4.6.2.5	16	---	---
Canister air flow resistance	3.4.1	4.6.2.8	100	25	---
Aerosol (DOP) penetrations before rough handling	3.4.2	4.6.2.1	25	---	---
Aerosol (DOP) pentrations after rough handling	3.4.3	4.6.2.2 and 4.6.2.3	---	25	---
Volatile content of canister	3.4.6	4.6.2.6	16	---	---

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TABLE II. First article inspection (destructive tests). - Continued

Item or characteristics	Requirement paragraph	Test paragraph	Number of canisters before rough handling	Number of canisters after rough handling	Number of individual canister parts
Canister aging	3.4.7	4.6.2.7	50	---	---
CK gas life aged 1/	3.4.4.1	4.6.2.4.1	25 of the canister aging samples	---	---
GB gas life before rough handling aged 1/	3.4.4.2	4.6.2.4.2	Remaining 25 of the canister aging samples	---	---
CK gas life unaged 2/	3.4.4.1	4.6.2.4.1	25	---	---
GB gas life before rough handling unaged 2/	3.4.4.2	4.6.2.4.2	25	---	---
GB gas life after rough handling 1/	3.4.4.3	4.6.2.4.3	---	50	---

1/ Samples shall be forwarded after aging in accordance with 4.6.2.7 to the Government test agency as specified in the contract or order (see 6.2.1).
2/ Samples shall be forwarded to the Government test agency as specified in the contract or order (see 6.2.1).

4.3.2.3 Acceptance/rejection criteria. If any one canister or component fails to meet 4.3.2.1 and 4.3.2.2, or if the unaged or aged gas life as determined by mean minus 1.1 times the standard deviation for each of the two gases, CK and GB, is less than the requirements of 3.4.4, where the standard deviation is as calculated in 6.7, the first article lot shall be rejected. No regular production lot shall be accepted prior to approval of a first article lot. The contractor shall obtain written approval from the contracting officer before proceeding with regular production. The contractor shall prepare a first article test report in accordance with the data ordering document included in the contract or order (see 6.2.2).

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the visual and nondestructive examination specified in 4.5 and group A, B, C and D tests specified in 4.4.3, 4.4.4, 4.4.5 and 4.4.6. The contractor shall prepare a test report in accordance with the data ordering document included in the contract or order (see 6.2.2). The contractor shall prepare a certificate of compliance in accordance with the data ordering document included in the contract or order (see 6.2.2) to assure that materials used comply with 3.2.

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4.4.1 Lot. A lot shall consist of the canisters produced by one manufacturer, at one plant, from the same materials under essentially the same manufacturing conditions during 1 week's production.

4.4.2 Sampling for quality conformance inspection.

4.4.2.1 Sampling for visual and dimensional examination. A random sample of parts, subassemblies, and assembled canisters shall be selected from each lot in accordance with MIL-STD-105, inspection level II, for the examination specified in 4.5. The Acceptable Quality Level (AQL) shall be 1.0 for major defects and 2.5 for minor defects. Any item in the sample containing one or more defects shall not be offered for delivery and if the number of nonconforming items in any sample exceeds the acceptance number for that sample, this shall be cause for rejection of the entire lot. If any sample contains a critical defect, the entire lot shall be rejected.

4.4.2.2 Sampling for group A tests (non-destructive). Each canister shall be subjected to the group A tests specified in 4.4.3. If any sample fails the group A tests, the entire lot shall be rejected.

4.4.2.3 Sampling for group B tests (destructive rubber). From every batch of rubber, under the observation of a Government inspector, test slabs shall be prepared for group B tests specified in 4.4.4. Test slabs shall be prepared using a cure equivalent to the production cure used to manufacture the rubber apron. The test slab size shall be as follows:

(a) Ten test slabs each 6 by 6 by 0.080 $\pm$ 0.005 inch.
(b) Two test slabs each 4 by 1.5 by 0.5 $\pm$ 0.005 inch.

If any sample fails the group B tests, the entire lot shall be rejected.

4.4.2.4 Sampling for group C tests (destructive). A random sample of parts, subassemblies, and assembled canisters shall be selected from each lot in accordance with MIL-STD-105 inspection level S-3, AQL 1.0, and tested as specified in 4.4.5. Any item in the sample containing one or more defects shall not be offered for delivery. If the number of nonconforming items in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.4.2.5 Sampling for group D tests (gas life). A random sample of assembled canisters shall be selected from each lot in accordance with MIL-STD-105 inspection level S-4, AQL 1.0, and tested as specified in 4.4.6. Any item in the sample containing one or more defects shall not be offered for delivery. If the number of nonconforming items in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.4.3 Group A tests. Each canister shall be subjected to the tests specified in table III.

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TABLE III. Group A tests.

Test	Requirement paragraph	Test paragraph
Air flow resistance of canister	3.4.1	4.6.2.8
Canister leakage	3.4.2	4.6.2.1
Packaging of canister (hermetic seal)	5.1.2	4.6.2.9

4.4.4 Group B tests. Group B tests are rubber destructive tests and sample test slabs shall be tested in accordance with 4.6.1.2.

4.4.5 Group C tests. Each sample part, subassembly, and assembly selected in accordance with 4.4.2.4 shall be subjected to the tests shown in table IV.

TABLE IV. Group C tests.

Test	Requirement paragraph	Test paragraph
Canister apron staking	3.4.5	4.6.2.5
Adsorbent volume	3.2.4	4.6.1.4.1
Adsorbent sieve size	3.2.4	4.6.1.4.2
Volatile content of canister	3.4.6	4.6.2.6
DOP, after rough handling	3.4.3	4.6.2.2, 4.6.2.3

4.4.6 Group D tests. Each assembly selected in accordance with 4.4.2.5 shall be subjected to the tests shown in table V. Sample containers shall be forwarded to the Government test agency specified in the contract or order (see 6.2.1).

TABLE V. Group D tests.

Test	Requirement paragraph	Test paragraph
CK gas life	3.4.4.1	4.6.2.4.1
GB gas life	3.4.4.2	4.6.2.4.2
GB gas life after rough handling	3.4.4.3	4.6.2.4.3

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4.5 Visual and nondestructive examination. Each sample part, subassembly, and assembled canister selected in accordance with 4.4.2.1 shall be examined for conformance to the requirements of table VI.

TABLE VI. Classification of defects.

a. Canister, C-1, Chemical-Biological, Mask, Mark V	
Categories	Defects
Critical	No defects permitted
1	Air flow resistance not as specified (see 3.4.1).
2	DOP penetration before rough handling not as specified (see 3.4.2).
3	Resistance mismatch (see 3.4.1).
Major	AQL 1.0
101	Rubber materials not free from pits, bumps, nonfills, porous areas, blisters, spongy areas, air traps, deformation flow cracks (see 3.6), foreign matter, flash abrasions, checks, tears, rips, cuts, holes, or punctures.
102	Canister improperly assembled (see 3.3.2).
Minor	AQL 2.5
201	Metal material or finish of metal surfaces not as specified on Drawing 803-1327146.
202	Color of rubber not as required (see 3.2.1.4).
203	Blooming of rubber not as specified (see 3.2.1).
204	Workmanship, other than 101.
205	Marking, characteristic data missing, incorrect, not permanent, or not legible (see 3.5).

b. Packaging inspection	
Categories	Defects
Critical	No defects permitted
1	Container leakage (5.1.2).
Major	AQL 1.0
101	Unit pack assembly incorrect.
102	Unit pack assembly not supplied with tear strip.
103	Corrosion resistant key not furnished.
104	Packing not as specified (see 5.1, 5.2, and 5.3).
105	Unit pack marking incorrect or illegible (see 5.4).
106	Packaging and packing (prior to shipping) not as required.

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4.6 Tests.

4.6.1 Material tests.

4.6.1.1 Filter paper additives. The contractor shall certify that the completed particulate filters do not contain a greater percentage of additives than is permitted by 3.2.2.

4.6.1.2 Rubber test methods. Each batch of rubber shall be evaluated for compliance with the specified physical property requirements of 3.2.1 and table I as specified in 4.6.1.2.1 through 4.6.1.2.7.

4.6.1.2.1 Aging. The aging shall be accomplished by exposing specimens of the rubber to oxygen at 158 ± 2°F and 300 ± 10 pounds per square inch (lb/in^2) for 48 hours. Testing shall be in accordance with ASTM D 573.

4.6.1.2.2 Tensile strength. Tensile strength shall be determined in accordance with ASTM D 412, method A. The sample shall be die size C and dumbbell shaped.

4.6.1.2.3 Elongation. Elongation shall be determined in accordance with ASTM D 412, method A. The sample shall be die size C and dumbbell shaped.

4.6.1.2.4 Tension set. Tension set shall be determined in accordance with ASTM D 412, method A. The sample shall be die size C and dumbbell shaped.

4.6.1.2.5 Hardness. Hardness shall be determined in accordance with ASTM D 1415, micro test, size 0.47 by 0.16 by 0.08 inch (12 by 4 by 2 mm). A stack of two samples is permitted.

4.6.1.2.6 Compression set. Compression set shall be determined by ASTM D 395, method B. Samples are to be clamped to 40 percent deflection and aged for 46 ± 1/4 hours at 90 ± 1°C (194 ± 1.8°F). Sample dimensions shall be:

(a) Diameter - 1.14 ± 0.02 inch (29.0 ± 0.5 mm).
(b) Thickness - 0.49 ± 0.02 inch (12.5 ± 0.5 mm).

4.6.1.2.7 Ozone resistance. The rubber compound for the canister aprons shall be tested for ozone resistance in accordance with ASTM D 1149. The test chamber shall be maintained at 104 ± 2°F (40 ± 1°C). The samples shall be exposed for 48 hours. At the end of this period, the aged samples, still stretched, shall be compared visually with a sample from the same test slab, similarly stretched but not aged. This comparison shall be made with the unaided eye. At least five samples shall be tested.

4.6.1.3 Cloth flannel disk. The air flow resistance of the cloth disk in the canister shall be determined at a linear air flow rate of 7.3 centimeters per second for conformance with 3.2.3 using Q112 Canister Air Purifier and Chemical Resistance Indicator in accordance with Drawing DLB31-6-351, or a Government approved commercial equivalent (see 6.9).

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4.6.1.4 Adsorbent.

4.6.1.4.1 Adsorbent volume. The volume of the adsorbent in each sample canister shall be determined using Q23 Chemical Container Volume Meter in accordance with Drawing E136-16-3, or a Government approved commercial equivalent (see 6.9). The adsorbent volume shall conform to the requirements of 3.2.4.

4.6.1.4.2 Adsorbent sieve size. The adsorbent in each sample from 4.6.1.4.1 shall be tested in accordance with the particle size test of MIL-C-13724 to determine conformance with 3.2.4.

4.6.2 Canister performance tests.

4.6.2.1 DOP smoke penetration. The DOP smoke penetration of each canister shall be determined using Q127 Penetrometer in accordance with Drawing DLB136-42-102, or Government approved commercial equivalent (see 6.9). The canister shall be tested at a continuous flow rate of 16 L/min for conformance with 3.4.2 and 3.4.3.

4.6.2.2 Rough handling (horizontal) before DOP smoke penetration. The sample canisters shall be tapped ten times in a horizontal position in accordance with Drawing 07609-65836 and shall then be tested in accordance with 4.6.2.1 for conformance with 3.4.3 using Q127 Penetrometer in accordance with Drawing DLB136-42-102, or Government approved commercial equivalent (see 6.9).

4.6.2.3 Rough handling (vertical) before DOP smoke penetration. The sample canisters shall be tapped ten times in a vertical position in accordance with Drawing 07609-65836 and shall then be tested in accordance with 4.6.2.1 for conformance with 3.4.3 using Q127 Penetrometer in accordance with Drawing DLB136-42-102, or Government approved commercial equivalent (see 6.9).

4.6.2.4 Gas life.

4.6.2.4.1 Cyanogen chloride (CK). The canister shall be equilibrated for 24 hours at 80 $\pm$ 3 percent relative humidity and 80 $\pm$ 10°F and then tested for cyanogen chloride gas life using the Q95 All Purpose Gas Life Testing Apparatus in accordance with Drawing DLB136-41-600. The canister shall be mounted so that the axis of flow is in a horizontal direction.

4.6.2.4.2 Sarin (GB). The canister shall be equilibrated for 24 hours at not more than 15 percent relative humidity and 75 $\pm$ 5°F and then tested for the GB gas life using the Q224, Tester, Filter Life, 50 L/min in accordance with Drawing DLB136-41-1800-5, or a Government approved commercial equivalent (see 6.9). The canister shall be tested singly mounted so that the axis of flow is in a horizontal direction.

4.6.2.4.3 Rough handling before Sarin (GB). No more than four sample containers, each containing two complete canisters, shall be rough handled at one time for 10 minutes in the Q113 Rough Handling Machine in accordance with Drawing C26-19-51, or a Government approved commercial equivalent (see 6.9). For samples less than four containers in number, dummy containers of equal weight shall be added to make up the difference.

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4.6.2.5 Canister apron staking. The staking of the canister apron shall be determined with the Q22 Canister Apron Pull Test Device in accordance with Drawing E136-13-362, or a Government approved commercial equivalent (see 6.9). The staking shall conform to the requirements of 3.4.5.

4.6.2.6 Adsorbent volatile matter. Sample canisters of the same lot which have been tested in accordance with 4.6.2.9 shall be divided into groups of two. The two canisters from a single hermetically sealed container shall constitute a group. Whenever there is an odd number of samples, one extra sample canister shall be selected. The adsorbent from each group of two sample canisters shall be completely removed and thoroughly blended to form a composite sample. Care shall be taken that the adsorbent is not exposed unnecessarily to moisture or rough handling during this blending operation. Two 10 $\pm$ 0.5 gram samples of adsorbent from each composite sample shall be weighed accurately. The moisture content of the samples shall be determined in accordance with ASTM D 2867. The volatile content shall be calculated on the dry weight of the adsorbent and shall conform to 3.4.6.

4.6.2.7 Aging. Sample containers containing two complete canisters shall be aged continuously for 14 days (336 hours) at 160 $\pm$ 5°F. The aged canisters shall be subjected to DOP penetration and gas life testing as specified in 3.4.7.

4.6.2.8 Air flow resistance of canister. The resistance of a canister and a particulate filter unit to a continuous air flow rate of 42.5 L/min (inhalation) shall be determined for conformance with 3.4.1 using the Q112 Canister Air Purifier and Chemical Container Resistance Indicator in accordance with Drawing DLB31-6-351, or a Government approved commercial equivalent (see 6.9). (Caution: The adsorbent in this canister will pick up moisture from the atmosphere; duration of test shall be as short as practical.)

4.6.2.9 Packaging of canister (hermetic seal). All sealed metal containers shall be tested by submerging the container so that the top is 1 inch below the surface of the water and held at 185 $\pm$ 5°F for 30 seconds to determine conformance with 5.1.2. Containers showing evidence of bubbles being released shall be rejected.

4.7 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.4.)

5.1 Unit pack. Unit packs shall be level A.

5.1.1 Level A. Unit packing shall be in accordance with Packaging Data Sheet (PDS) 5-3-578 except that the can will be finished gray No. 36307 of FED-STD-595. The resulting unit pack is one hermetically sealed can in accordance with PPP-C-96 (the unit of issue) containing two canisters.

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5.1.2 Unit container leakage. The unit container shall show no sign of leakage as evidenced by release of a continuous stream of bubbles which appear at any surface when tested as specified in 4.6.2.9.

5.2 Intermediate pack. Twenty-four unit packs shall be placed in a PPP-B-636 W5C fiberboard container in accordance with the instructions in PDS 5-3-578.

5.3 Packing. Packing shall be level A, B, or C, as specified (see 6.2.1). Use of polystyrene, loose fill material is prohibited for all levels of packing. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation. Each pack shall contain a completed Material Safety Data Sheet.

5.3.1 Level A. Not more than 30 intermediate packs (720 unit packs) shall be packed in an exterior pack constructed in accordance with PPP-B-601. Boxes weighing over 200 pounds shall be skidded.

5.3.2 Level B. Not more than 30 intermediate packs (720 unit packs) shall be packed in an exterior container constructed in accordance with PPP-B-640, Class 2. Boxes weighing over 200 pounds shall be skidded.

5.3.3 Level C. Not more than 30 intermediate packs shall be stretch-wrapped on an expendable pallet.

5.4 Marking. Marking shall be in accordance with MIL-STD-129. Nomenclature shall be "Canister, C-1, for Mask, Chemical-Biological, MK V". Lot numbers are required. Unit packs shall, in addition to identification marking, be marked with the statement, "This sealed, airtight can contains two canisters. Do not open until required for use on the MK V Mask. To remove, open can with key attached to end of can. Warning: Do not use this can as a food or drinking container (POISON)."

6. NOTES

6.1 Intended use. The C-1 canister is intended to be used on the MK V Chemical-Biological Mask. This canister is not to be used for protection against gasoline vapors, refrigerants, carbon monoxide fumes, and other industrial gases.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) If first article inspection is required (see 3.1).
(c) Name and address of test facility and shipping instructions for gas life tests (see 4.3.2.2, table II and 4.4.6).
(d) Level of packing (see 5.3).

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6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of FAR 52.227-7031 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
3.7	Manufacturing methods report	DI-P-1604	---
4.1.1	Inspection system program plan	DI-R-4803	---
4.1.2	Toxicity report, protective masks	DI-T-1911	---
4.3.2.3	First article inspection report	DI-T-4902	---
4.4	Reports, test	DI-T-2072	10.1.b
4.4	Certification data/report	UDI-A-23264	---

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.2.3 The contracting officer should attach a copy of Drawing TDPL 5-3-578 to the contract for information purposes.

6.2.4 Toxicity data. The information listed below must be submitted to the contracting officer with the bid submittal to establish the nontoxicity of the formulation. The US Army Surgeon General will act as advisor to the contracting officer in evaluating this data for determining if the formulation and parts are acceptable. The information relating to the description of the candidate formulations may be classified as "company confidential", in which case it will be so treated by the Government.

(a) Information relating to candidate formulations:

(1) The name of the contractor and the producing subcontractor, if any.

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(2) Description of the item and end item and copies of applicable specifications shall be included.
(3) The number of items under acquisition for which the proposed formulation shall be used.
(4) A description of the candidate formulation shall include:

a. The chemical and trade names of the ingredients.
b. The quantity of each ingredient, including the range of variation that can be tolerated without loss of quality in the finished item. Quantities shall be expressed in percent by weight and parts by weight.
c. The proposed mold lubricants, dusting powders, wet spray, or other materials used in the manufacturing process.
d. Proposed cure times and temperatures.
e. Results of previous toxicity evaluation demonstrating acceptability or evidence of prior Government approval of the proposed formulation.
f. Information and data to substantiate the nontoxicity of the product. This information shall include a complete description of any programs the contractor has carried out to prove nontoxicity, such as prior patch testing with unlined items the use of which involves direct contact with the skin; evidence that the proposed formulation is the same as that used in his commercial products; and information on the nature, extent, and length of time of his commercial experience with the formulation. Other evidence may include references to technical or other scientific publications commercial laboratory reports, opinions or statements of recognized authorities and agencies in the field of dermatology, and any other pertinent data.

6.3 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 Canister closing. The assembly line should be so arranged that the canister, after filling, is exposed to the atmosphere for the shortest possible time. Immediately after filling and closing, the canister should be sealed by a suitable closure, placed in an airtight container such as a charcoal drum, or other adequate containers, to prevent undue moisture adsorption, pending placement of the canister in the appropriate packaging container. These precautions are necessary because charcoal can adsorb sufficient moisture during the manufacturing operations to cause the canister to be rejected.

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6.6 Major change. The term applies to any change in formulae, processing, or curing of currently approved rubber stocks which induce toxic characteristics and cause casualties. The major change should be considered effected when one or more of the following conditions occur:

(a) Components of formulae are varied in excess of the limitations stated in the manufacturer's tolerance (see 4.1.2).
(b) The addition or deletion of one or more components of formulae.
(c) Substitutes for one or more components of formulae without advance approval of the Government.
(d) Deviation from the submitted, approved factory mold cure temperature exceeding plus 5°F or minus 10°F even though equivalent cure is attained.
(e) Use of mold lubricants, dusting powders, or wet sprays not submitted with original information for toxicity approval.

6.7 Standard deviation. To find the standard deviations of a group of results:

Assume $x_1, x_2, \ldots, x_n$ constitute numerical information recorded from a group of n results. (x_i will denote the numerical value recorded for the ith object in the sample.)

Step 1. Find the mean, $\bar{x}$

$$\bar{x} = \sum_{i=1}^{n} x_i/n$$

Step 2. Find the variance, s^2

$$s^2 = \sum_{i=1}^{n} (x_i - \bar{x})^2/(n - 1)$$

Step 3. Find the standard deviation, s

$$s = \sqrt{s^2}$$

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Alternate method (for use on computer):

Step 1. Find the mean, $\bar{x}$

$$\bar{x} = \sum_{i=1}^{n} x_i/n$$

Step 2. Find the variance, s^2

$$s^2 = (\sum_{i=1}^{n} x_i^2 - n\bar{x}^2)/(n - 1)$$

Step 3. Find the standard deviation, s

$$s = \sqrt{s^2}$$

6.8 Charcoal storage. After opening a container of charcoal, it is recommended that the container, with the unused quantity of charcoal, be stored at a temperature of 70 ± 5°F and a relative humidity of 30 ± 5 percent.

6.9 Test equipment. The Government agency responsible for approval of commercial equivalent test equipment is the US Army Armament, Munitions and Chemical Command, AMSMC-QAC-P, Aberdeen Proving Ground, Edgewood, MD 21010-5423.

6.10 Rubber compounds. The Government agency responsible for approval of commercial equivalent rubber compound additives is the Naval Sea Systems Command, Personnel Protection Branch, Washington, DC 20362-5101.

6.11 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 4240-N814)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-24314C(SH)	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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